Coding Redundancy

Coding redundancy is a type of redundancy in image compression where **more bits than necessary are used to represent image data**.

The main goal is to reduce the number of bits required to store or transmit an image.

Simple Definition

If the same information can be represented using fewer bits, then the extra bits are called **coding redundancy**.

Example

Suppose an image contains pixels:

A A A A A B B C

If we store every symbol using fixed 3 bits:

Symbol	Code
A	001
B	010
C	011

Then:

A A A A A B B C
= 001 001 001 001 001 010 010 011

Many bits are used.

But since A appears most frequently, we can use shorter codes:

Symbol	Variable Length Code
A	1
B	01
C	001

Now fewer bits are needed.

This reduction in unnecessary bits is called **removing coding redundancy**.

Main Idea

- Frequently occurring symbols → shorter codes
- Rare symbols → longer codes

This technique reduces total storage size.

Formula

Coding redundancy is related to:

- Average bits used per symbol
- Minimum possible bits (entropy)

The closer the coding is to entropy, the better the compression.

Techniques Used

1. Huffman Coding

- Most common method
- Uses variable-length codes
- Frequently occurring data gets shorter codes

2. Arithmetic Coding

- Represents whole message as a single fraction value
- Gives better compression than Huffman in some cases

Advantages

- Reduces image size
- Saves storage
- Faster transmission
- Lossless compression possible

Disadvantages

- More complex encoding/decoding
- Variable-length decoding may be slower

Real-Life Example

ZIP files, PNG images, and JPEG compression all use coding redundancy reduction techniques.

One-Line Interview Answer

Coding redundancy occurs when image data is represented using more bits than necessary. It is reduced using efficient coding methods like Huffman coding and Arithmetic coding to achieve image compression.

Interpixel Redundancy

Interpixel redundancy is a type of redundancy in images where **neighboring pixels contain similar or repeated information**.

In most images, adjacent pixels are highly correlated, meaning nearby pixels usually have similar intensity or color values.

Simple Definition

When neighboring pixels carry almost the same information, the repeated information is called **interpixel redundancy**.

Example

Consider pixel values:

52 52 52 53 53 52

Here, most neighboring pixels are nearly identical.

Instead of storing every value separately, we can store:

- one original value
- and the difference between adjacent pixels

Differences:

52 0 0 1 0 -1

This requires less data for compression.

Main Idea

Images usually contain:

- smooth areas
- repeated patterns
- gradual intensity changes

So storing every pixel independently wastes space.

How It Is Reduced

Interpixel redundancy is reduced using techniques like:

1. Run-Length Encoding (RLE)

Stores repeated pixels as:

(value, count)

Example:

52 repeated 5 times → (52,5)

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2. Predictive Coding

Predicts current pixel from neighboring pixels and stores only the error.

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3. Differential Encoding

Stores differences between adjacent pixels instead of actual values.

Advantages

- Reduces storage size
- Improves compression efficiency
- Used in many image/video compression systems

Applications

Used in:

- JPEG compression
- PNG compression
- Video compression (MPEG, H.264)

One-Line Interview Answer

Interpixel redundancy occurs because neighboring pixels in an image are highly correlated and contain similar information. It is reduced using predictive coding, run-length encoding, and differential encoding techniques.

Psychovisual Redundancy

Psychovisual redundancy refers to the image information that the **human eye cannot easily notice or is less sensitive to**.

During image compression, this less important visual information can be removed without causing major visible changes in the image.

Simple Definition

The human visual system does not notice all image details equally.
Removing less noticeable details is called **psychovisual redundancy reduction**.

Main Idea

Human eyes are:

- more sensitive to brightness than color
- less sensitive to very fine details
- unable to notice small changes in high-frequency components

So these less important details can be compressed or discarded.

Example

Suppose two images look almost identical to human eyes:

- Original image → 10 MB
- Compressed image → 2 MB

Even though some data is removed, humans may not notice the difference.

That removed unnecessary visual information is psychovisual redundancy.

How It Is Reduced

1. Quantization

Less important frequency components are reduced or removed.

Used heavily in JPEG compression.

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2. Color Subsampling

Human eyes are less sensitive to color changes than brightness.

So color information is compressed more aggressively.

Example:

4:2:0 chroma subsampling

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3. Removing High-Frequency Details

Tiny details and sharp variations may be discarded.

Used In

- JPEG images
- MPEG videos
- MP3 audio compression (similar concept)

Advantages

- Achieves very high compression ratio
- Saves storage and bandwidth
- Maintains visually acceptable quality

Disadvantages

- Lossy compression
- Some image details are permanently lost

One-Line Interview Answer

Psychovisual redundancy is the visually insignificant information in an image that human eyes cannot easily perceive. Removing this information helps achieve high image compression with minimal visible quality loss.

Huffman Coding

Huffman Coding is a **lossless compression technique** used to reduce the size of data by assigning:

- **shorter codes** to frequently occurring symbols
- **longer codes** to less frequent symbols

It is widely used in image compression, file compression, and JPEG.

Simple Definition

Huffman Coding compresses data by using variable-length binary codes based on symbol frequency.

Main Idea

If a symbol appears many times, it should use fewer bits.

Example:

Symbol	Frequency	Huffman Code
A	50	0
B	30	10
C	15	110
D	5	111

Here:

- A is most frequent → shortest code
- D is least frequent → longest code

Steps of Huffman Coding

Step 1: Calculate Frequency

Count how many times each symbol appears.

Example:

Symbol	Frequency
A	5
B	9
C	12
D	13
E	16
F	45

Step 2: Create Nodes

Create one node for each symbol.

Step 3: Build Huffman Tree

- Select two smallest frequencies
- Merge them
- Repeat until one tree remains

Step 4: Assign Binary Codes

- Left branch → 0
- Right branch → 1

Generate unique binary codes.

Huffman Tree Example

```
     (100)
    /    \
  F(45)   (55)
         /   \
      (25)   (30)
```

Codes are generated from root to leaf.

Properties

- Variable-length coding
- Prefix code property:
    - no code is the prefix of another code
- Lossless compression

Advantages

- Reduces storage size
- No data loss
- Efficient for repeated patterns
- Simple and widely used

Disadvantages

- Requires frequency table
- Less effective if frequencies are similar
- Huffman tree must be stored/transmitted

Applications

Used in:

- JPEG
- PNG
- ZIP files
- Text compression

One-Line Interview Answer

Huffman Coding is a lossless compression technique that assigns shorter binary codes to frequently occurring symbols and longer codes to less frequent symbols to reduce storage size efficiently.

Arithmetic Coding

Arithmetic Coding is a **lossless data compression technique** in which the entire message is represented by **a single fractional number between 0 and 1**.

It achieves better compression efficiency than Huffman Coding in many cases.

Simple Definition

Instead of assigning separate binary codes to each symbol, Arithmetic Coding encodes the whole message into one decimal/fraction value.

Main Idea

- Symbols with higher probability get smaller ranges
- The message is repeatedly narrowed into a smaller interval
- Final interval represents the complete data

Example

Suppose probabilities are:

Symbol	Probability
A	0.5
B	0.3
C	0.2

Initial range:

0.0 to 1.0

Assign ranges:

Symbol	Range
A	0.0 – 0.5
B	0.5 – 0.8
C	0.8 – 1.0

If message = "AB":

1. Start with:

[0,1]

2. After A:

[0,0.5]

3. After B:
   Take B’s portion inside [0,0.5]

Final range:

[0.25,0.40]

Any number inside this interval can represent "AB".

Important Concept

The longer the message, the smaller the final interval becomes.

That final fraction uniquely identifies the data.

Advantages

- Better compression than Huffman coding
- Works well for skewed probabilities
- Very efficient for large data

Disadvantages

- More computationally complex
- Slower encoding/decoding
- Difficult implementation compared to Huffman coding

Applications

Used in:

- JPEG/JPEG2000

- Video compression
- Text compression
- CABAC in H.264/HEVC

Huffman vs Arithmetic Coding

Feature	**Huffman**	**Arithmetic**
Coding Method	Symbol-wise	Whole message
Compression Efficiency	Good	Better
Complexity	Simple	Complex
Speed	Faster	Slower

One-Line Interview Answer

Arithmetic Coding is a lossless compression technique that represents an entire message as a single fractional value between 0 and 1 using symbol probabilities for efficient compression.

Lossy Compression Techniques

Lossy compression is a compression method in which **some image data is permanently removed** to reduce file size.

After decompression, the reconstructed image is not exactly identical to the original image.

Simple Definition

Lossy compression removes less important image information to achieve high compression.

Main Idea

Human eyes cannot notice every small detail in an image.

So unnecessary or less visible information is discarded to save storage and bandwidth.

Characteristics

- High compression ratio
- Some quality loss occurs
- Irreversible process
- Smaller file size

Working Process

Typical steps in lossy image compression:

1. Transform image data
2. Remove less important information
3. Quantization
4. Encoding compressed data

Common Lossy Compression Techniques

1. Transform Coding

Image is converted from:

- spatial domain → frequency domain

Usually using:

- DCT (Discrete Cosine Transform)
- Wavelet Transform

High-frequency components are reduced because human eyes are less sensitive to them.

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2. Quantization

Most important step in lossy compression.

Small details are removed by reducing precision.

Example:

52.786 → 53

This causes data loss but reduces storage.

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3. Predictive Coding

Predicts pixel values using neighboring pixels and stores only prediction error.

Small errors require fewer bits.

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4. Vector Quantization

Groups similar image blocks together and stores representative patterns.

JPEG Compression (Most Common Example)

JPEG uses:

- DCT transform
- Quantization
- Huffman coding

It removes visually unimportant details to compress images efficiently.

Advantages

- Very small file size
- Faster transmission
- Saves storage space
- Suitable for multimedia

Disadvantages

- Image quality decreases
- Data cannot be perfectly recovered
- Repeated compression reduces quality further

Applications

Used in:

- JPEG images
- MP4 videos
- Streaming platforms
- Social media image uploads

Lossy vs Lossless Compression

Feature	Lossy	Lossless
Data Loss	Yes	No
File Size	Smaller	Larger
Quality	Reduced	Original preserved
Example	JPEG	PNG

JPEG Compression

JPEG (Joint Photographic Experts Group) is a widely used **lossy image compression technique** for reducing image file size while maintaining acceptable visual quality.

It is mainly used for:

- digital photos
- web images
- multimedia applications

Simple Definition

JPEG compression reduces image size by removing visually less important information from the image.

Main Idea

Human eyes:

- are less sensitive to high-frequency details
- notice brightness more than color variations

JPEG uses this property to compress images efficiently.

Steps of JPEG Compression

1. Color Space Conversion

The image is converted from:

RGB → YCbCr

Where:

- Y = brightness (luminance)
- Cb, Cr = color information (chrominance)

Human eyes are more sensitive to brightness than color.

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2. Chroma Subsampling

Color components are reduced because small color changes are less noticeable.

Example:

4:2:0 subsampling

This reduces data size.

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3. Divide Image into Blocks

The image is divided into:

8 × 8 pixel blocks

Each block is compressed separately.

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4. Apply DCT (Discrete Cosine Transform)

Each 8×8 block is converted from:

- spatial domain → frequency domain

DCT separates:

- low-frequency components
- high-frequency components

Low frequencies contain most visual information.

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5. Quantization

Most important lossy step.

High-frequency values are reduced or removed.

This decreases image quality slightly but greatly reduces size.

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6. Zig-Zag Scanning

DCT coefficients are arranged in zig-zag order:

- low frequencies first
- high frequencies later

This helps compression.

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7. Entropy Encoding

Finally compressed using:

- Huffman Coding
- Arithmetic Coding

JPEG Compression Flow

Image
→ Color Conversion
→ Chroma Subsampling
→ 8×8 Blocks
→ DCT
→ Quantization
→ Zig-Zag Scan
→ Huffman Coding
→ JPEG File

Advantages

- Very high compression ratio
- Small file size
- Good visual quality
- Widely supported

Disadvantages

- Lossy compression
- Quality decreases after repeated editing/saving
- Compression artifacts may appear

Applications

Used in:

- Digital cameras

- Websites
- Social media
- Image storage systems

One-Line Interview Answer

JPEG compression is a lossy image compression technique that reduces image size using DCT, quantization, and entropy coding while maintaining acceptable visual quality.